

# Ian Leung

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## EDUCATION

### University of Waterloo

Sep 2025 – Present

*Bachelor of Computer Science and Finance - Double Major*

- President's Scholarship of Distinction (\$5000), Undergraduate Achievement Award (\$1500)
- Statistics Minor

**Relevant Coursework:** Functional Programming (Advanced), Interpreters (Advanced), Linear Algebra, Calculus, Probability, Proofs, Financial Markets and Data Analytics

## TECHNICAL SKILLS

Languages Python, Java, C++, C#, C, Go, Kotlin, TypeScript, SQL, HTML, CSS, Bash  
Technologies Temporal, Pytorch, Supabase, Firebase, MongoDB, React, Maven, Qt, Openrouter, OpenCV  
Developer Tools Linux, Git, Docker, Jupyter, DigitalOcean, Modal, Render, Vercel, Unity, Android Studio

## WORK EXPERIENCE

### Roblox

San Mateo, CA

*Software Engineer Intern — Go, Python*

Jun 2026 – Present

- Developing an automated CI/CD pipeline for network configuration deployments using **GitHub Actions** and **Temporal**, replacing manual rack-switch config push workflows with a safer, auditable rollout system
- Designing a canary-first rollout strategy with pre/post-deployment health checks, progressive rollout gates, and staged batch expansion to safely validate and deliver network changes to **100+ million** daily active users
- Implementing deployment safety mechanisms including device locking, validation gates, and rollback logic to prevent unsafe configuration pushes during canary and batch rollouts

### Sunnybrook Research Institute

Toronto, ON

*Software Research Intern — Python*

Jul 2025 – Aug 2025

- Built an event tracing and simulation tool to simulate medical professionals' interactions with **Focused Ultrasound** 🔄 software, enabling realistic testing and diagnosis of software defects, unintended user behaviours, and edge cases
- Created an automated software testing framework with **PyQt** and **Pydantic**, generating synthetic and replayable JSON event flows, reducing software testing times by **91%**
- Implemented a logging API using **Difflib** and **Loguru** to validate test results against baselines, identifying **5+** bugs

### Data Science Club

Waterloo, ON

*Software Developer — TypeScript, HTML, CSS*

Sep 2025 – May 2026

- Maintaining [www.uwdatascience.ca](http://www.uwdatascience.ca) - UWDC's primary website and associated apps for events and internal tooling
- Created an internal onboarding form and API to manage executive profiles using **TypeScript**, **React** and **Supabase**

## PROJECTS

### 🔗 Pharmagotchi: Pharmacy Companion App | Android Studio, Figma, Kotlin, XML, OpenRouter

- Built an **Android app** featuring a virtual pharmacy companion that provides users with health tracking, medication reminders, AI-powered advice, and direct communication channels with healthcare providers
- Leveraged **OpenRouter LLMs** to integrate a virtual assistant for personalized health advice, symptom checking, and medication information, allowing users to better understand their health and medications
- Generates automated health reports and medication summaries for users to easily share with healthcare providers

### 🔗 PeerAssist - Full-Stack Peer Reviewing | Java, MongoDB, Apache PDFBox, Maven

- Created a full-stack **Java** desktop application for students to upload, review and grade peers' academic work
- Integrated a **Supabase** backend for authentication and data storage, enforcing **Row-Level Security** policies across profiles, documents, and reviews tables to control per-user access
- Rendered uploaded PDFs with **Apache PDFBox** and added search, sorting, and ranking by average review score; packaged the app into a single zero-config executable JAR via **Maven** shade plugin

### 🔗 Trust No Ghost - 3D Maze Exploration Game | Unity, C#, Itch.io

- Shipped Trust No Ghost 🔄 for **60+ users** at the UW Game Jam Fall 2025, placing in the **top 10** most rated games
- Utilized Unity and C# to develop expanding level designs, AI pathfinding algorithms, and dynamic moving mechanics